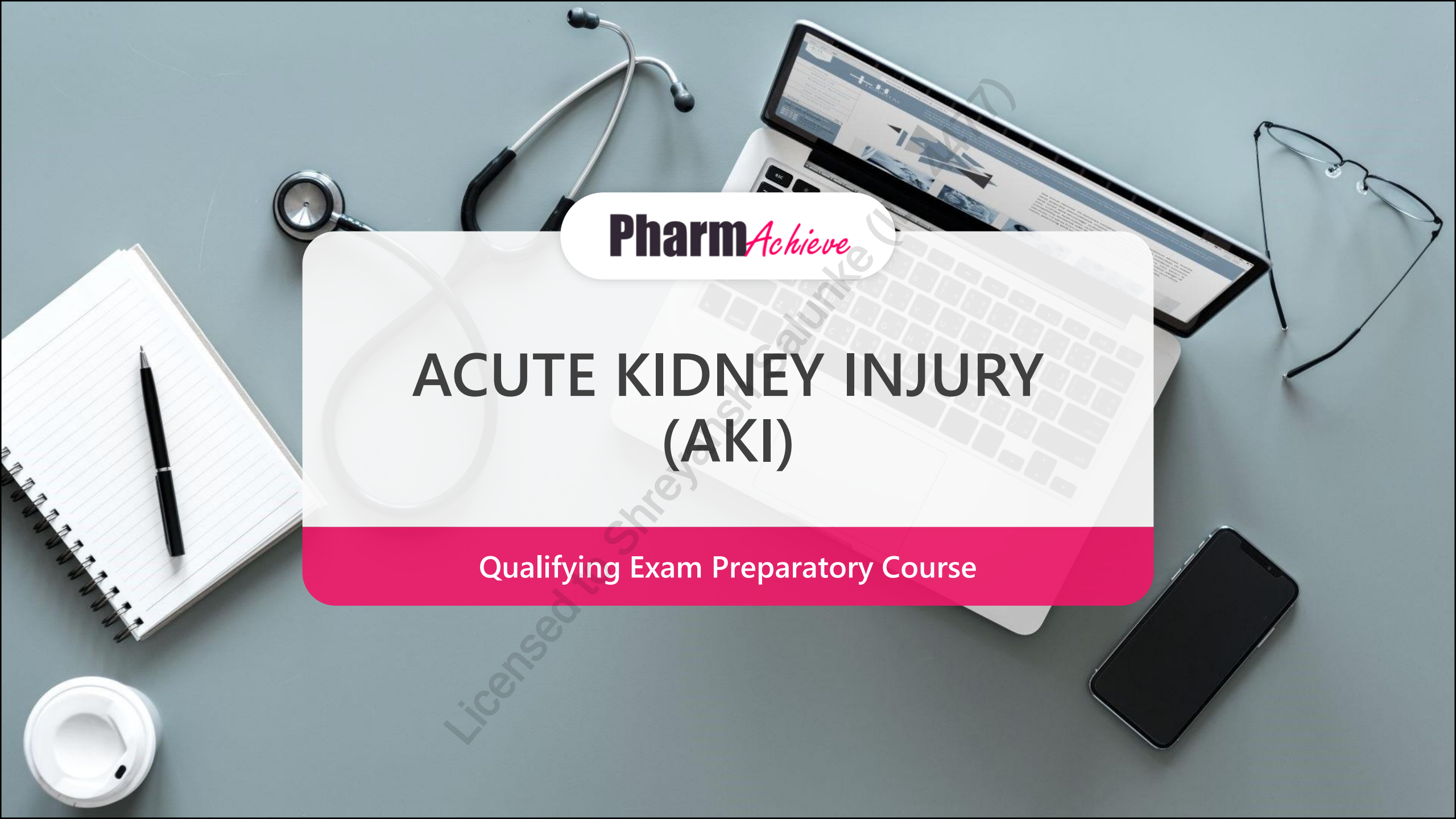


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

ACUTE KIDNEY INJURY (AKI)

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AdjBW Adjusted Body Weight

ABW Actual Body Weight

ACEi Angiotensin Converting Enzyme Inhibitor

ARB Angiotensin II Receptor Blocker

AKI Acute Kidney Injury

BP Blood Pressure

BUN Blood Urea Nitrogen

Ca²⁺ Calcium

CKD Chronic Kidney Disease

CrCl Creatinine Clearance

CVD Cardiovascular Disease

ECG Electrocardiogram

eGFR Estimated Glomerular Filtration Rate

GERD Gastroesophageal Reflux Disease

HR Heart rate

IBW Ideal Body Weight

IV Intravenous

JVP Jugular Venous Pressure

K⁺ Potassium

KDIGO Kidney Disease Improving Global Outcomes

Na⁺ Sodium

NS Normal Saline

NSAIDs Non-Steroidal Anti-Inflammatory Drugs

PO Orally

PO₄³⁻ Phosphate

RAAS Renin-Angiotensin-Aldosterone System

RBC Red Blood Cells

RRT Renal Replacement Therapy

SCr Serum Creatinine

TID Three Times Daily

TLS Tumor Lysis Syndrome

WBC White Blood Cells

OBJECTIVES

- 1 Define the three broad categories of acute kidney injury (pre-renal, intra-renal, post-renal)
- 2 Define the pathophysiology of acute kidney injury
- 3 Discuss the risk factors associated with acute kidney injury
- 4 Discuss various causes of acute kidney injury
- 5 Understand the risk of hyperkalemia and the treatment regimens
- 6 Define the monitoring parameters for acute kidney injury and the indications for renal replacement therapy (dialysis)

DEFINITIONS

Acute Kidney Injury (AKI)

- 2012 KDIGO consensus definition and criteria:
 - ↑ Serum creatinine (SCr) $\geq 26.5 \mu\text{mol/L}$ within 48 hours, OR
 - ↑ SCr $\geq 1.5\times$ baseline known/presumed to have occurred within the prior 7 days, OR
 - Urine volume $<0.5 \text{ mL/kg/hour}$ for 6 hours
- KDIGO criteria is also used to stage AKI (stages 1 through 3)
- May need renal replacement therapy (RRT)
- Mortality 25-80%

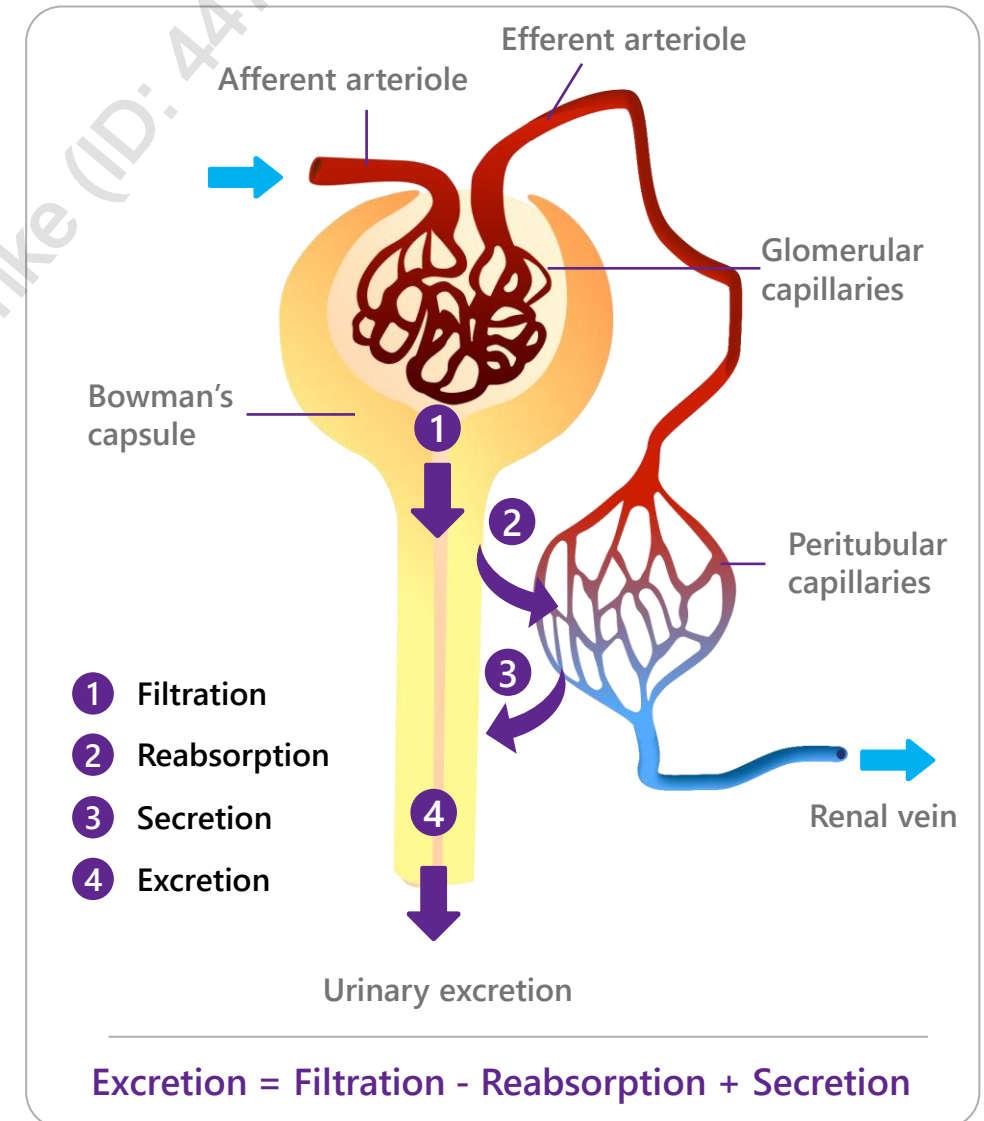
RISK FACTORS

- **Co-morbidities**
 - CVD, history of cardiac surgery
 - Diabetes
 - Multiple myeloma
 - Hepatorenal syndrome
 - Rhabdomyolysis
- **CKD**
- **Hypovolemia**
- **Medications/toxins, contrast dye**
- **Sepsis**
- **People of African descent**
- **Advanced age**
 - Due to increased risk for dehydration and renal atherosclerotic disease

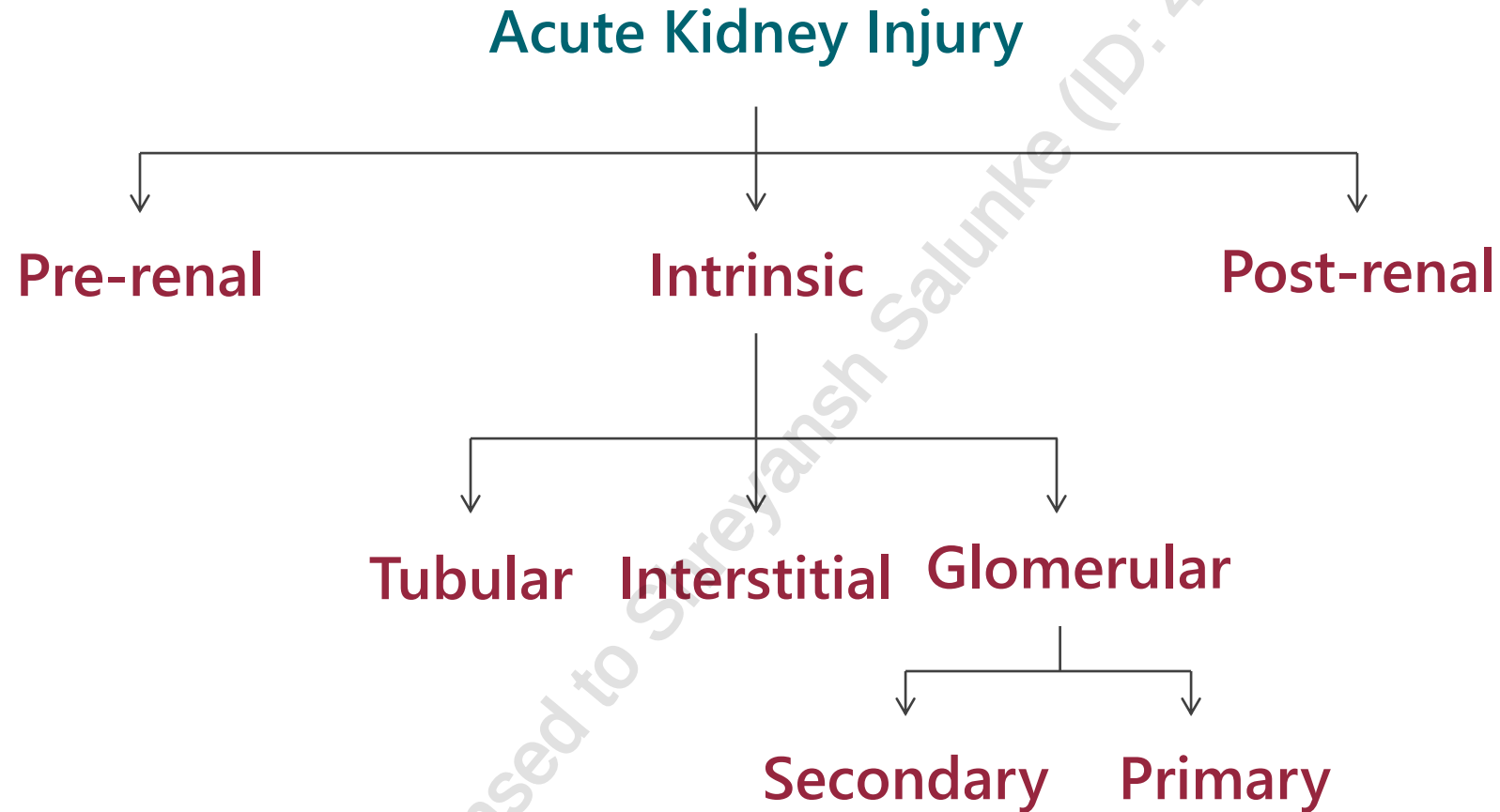
PATHOPHYSIOLOGY

Kidney Functions:

- ✓ Removes metabolic waste (creatinine, urea)
- ✓ Balances fluids
- ✓ Balances electrolytes (K^+ , PO_4^{3-} , Na^+ , Ca^{2+})
- ✓ Metabolizes vitamin D precursor to active vitamin D
- ✓ Maintains acid-base regulation
- ✓ Stimulates red blood cell production via erythropoietin



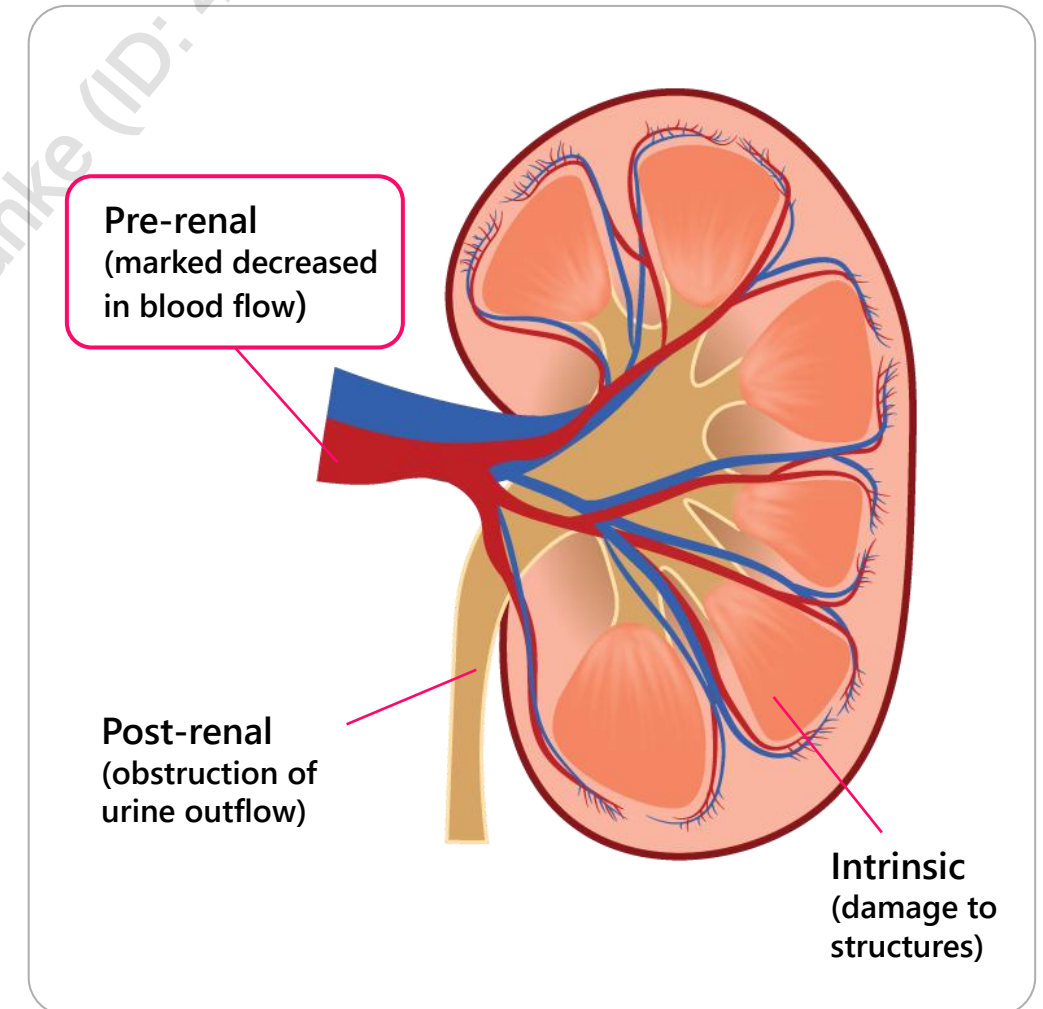
TYPES OF AKI



TYPES OF AKI

PRE-RENAL (FUNCTIONAL) – 70% of all cases

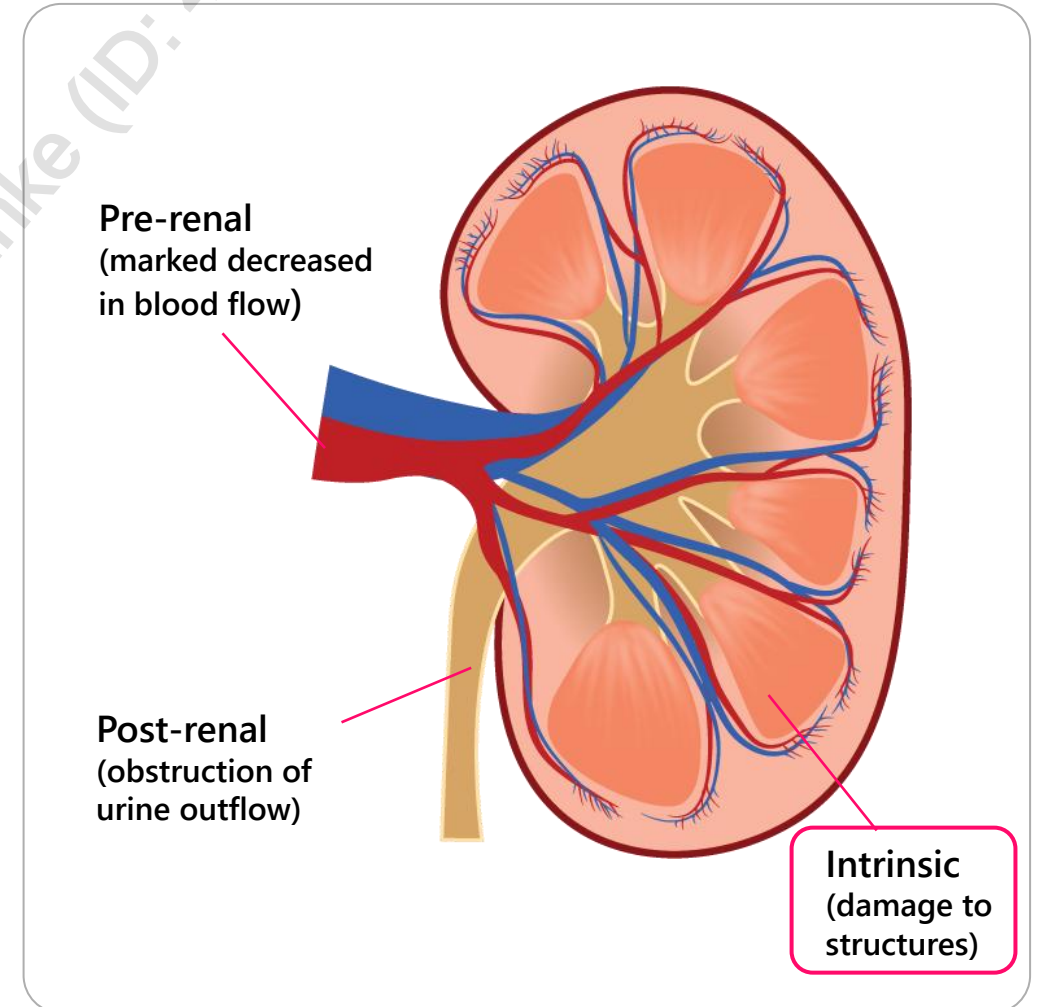
- **Hypo-perfusion** of kidney
- **BUN/SCr ratio increases**
- **Intrarenal vasoconstriction**
 - Medications, hepatorenal syndrome, cardiorenal syndrome, hypercalcemia
- **Systemic vasodilation**
 - Sepsis, neurogenic shock
- **Volume depletion**
 - Vomiting, diarrhea, sweating, blood loss (extra-renal loss)
 - Diuretic overuse, osmotic diuresis (renal loss), burns



TYPES OF AKI

INTRINSIC (STRUCTURAL) – 10% of all cases

- **Structural damage** due to **ischemia, inflammation, toxins, necrosis**
- **Tubular damage**
 - Prolonged ischemia, exogenous toxins (medications) and endogenous toxins (rhabdomyolysis, tumour lysis syndrome, myeloma)
- **Glomerular damage**
 - Acute inflammation of blood vessels and glomeruli
 - Glomerular nephritis
- **Interstitial damage**
 - **Most often medication-related**
 - Infection (bacterial, viral, fungal)
 - Systemic disease (sarcoidosis, lupus)
- **Vascular damage**
 - Renal vein thrombosis, renal infarction
 - Malignant hypertension, renal embolisms

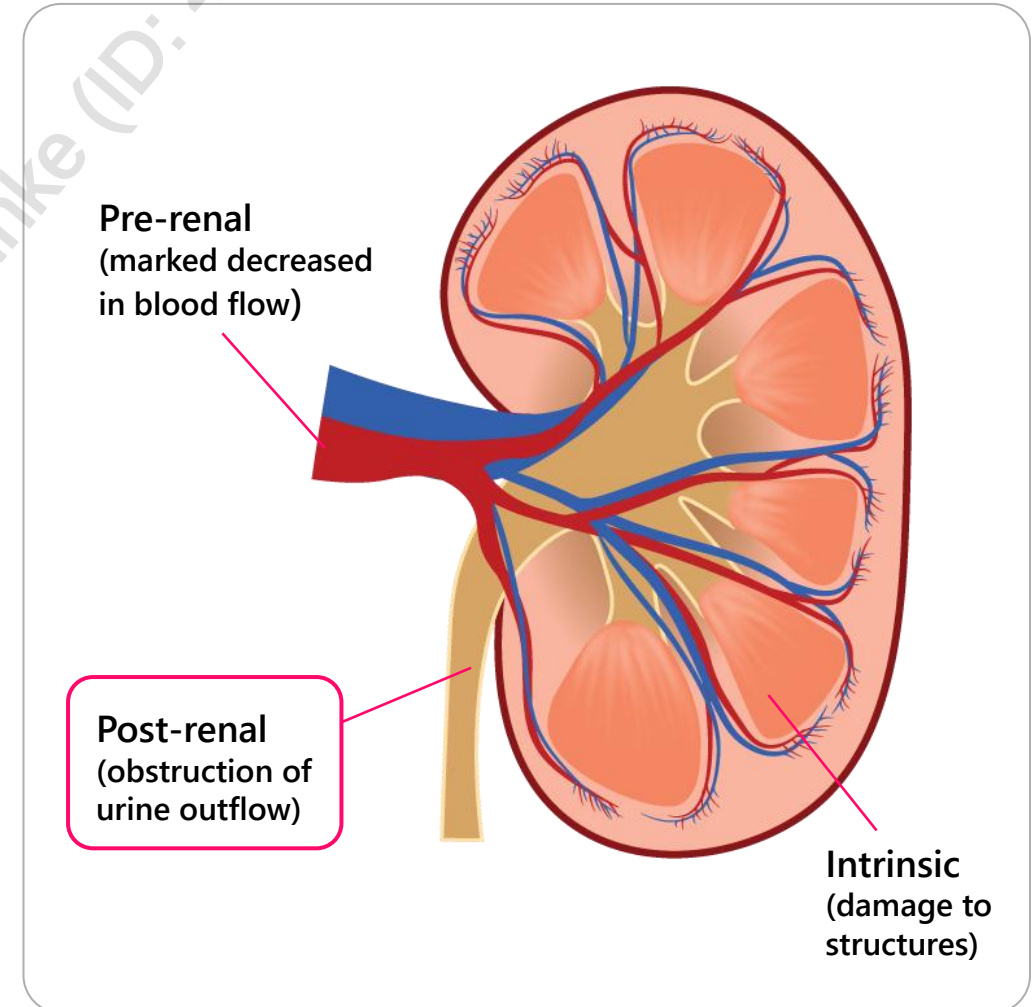


TYPES OF AKI

POST-RENAL– 20% of all cases

Obstruction to urine outflow (ureter)

- Medications
- Cancer
 - Bladder, prostate, cervical
- Benign prostatic hypertrophy
- Neurogenic bladder
 - Extrarenal obstruction
- Kidney stones
- Clots
- Tumors
 - Tumor lysis syndrome (TLS)



SUMMARY – TYPES OF AKI

**70% of all causes
HYPOPERFUSION OF THE
KIDNEYS**

- Intrarenal vasoconstriction
- Systemic vasodilation
- Volume depletion

**20% of all causes
OBSTRUCTION TO URINE
FLOW**

- Bladder obstruction
- Tumour lysis syndrome
- Clots
- Kidney stones

Pre-renal
(marked decreased
in blood flow)

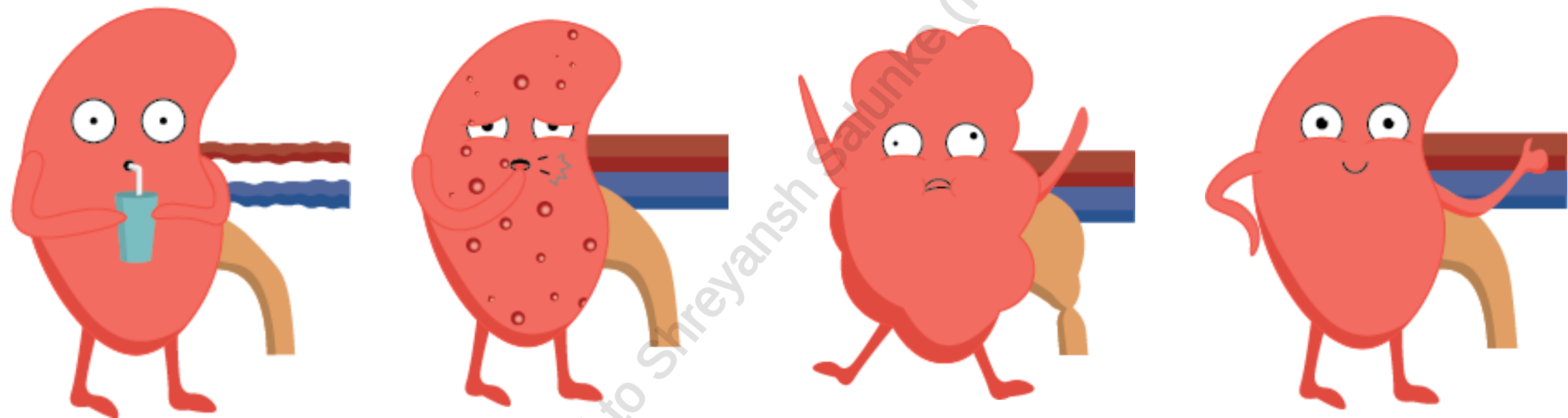
Post-renal
(obstruction of
urine outflow)

Intrinsic
(damage to
structures)

**10% of all causes
STRUCTURAL
DAMAGE DUE TO
ISCHEMIA,
INFLAMMATION,
TOXINS AND
NECROSIS**

- Tubular damage
- Glomerular damage
- Interstitial damage
- Vascular damage

LABEL THE AKI!



CLINICAL PRESENTATION

Mild-to-moderate AKI

- Mostly asymptomatic
- Usually detected through bloodwork

Severe AKI

- Restlessness
- Confusion
- Fatigue
- Anorexia
- Flank pain
- Weight gain or edema
- Oliguria → <400 mL urine/day
- Anuria → <100 mL urine/day
- Proteinuria
- Electrolyte disturbances
 - ↑K⁺
 - ↑PO₄³⁻
- Uremic encephalopathy

DIAGNOSIS

- SCr, Urea (BUN)
 - eGFR not reliable in AKI
 - Electrolytes
- Urine output
- Serum and urinary sodium
- Urine osmolality
- Urinalysis
 - Renal tubular cells/casts, eosinophils, WBC casts, RBC casts, protein
- Renal ultrasound
- Renal biopsy (if intrinsic damage is suspected)
- History of nephrotoxin use, possible sepsis or hypovolemia

PROGNOSIS

- Mortality - variable based on underlying disease and time to treatment initiation
 - Higher mortality
 - Elderly
 - Sepsis
 - Multi-organ failure
 - Hepatorenal dysfunction
- Risk factor for developing CKD

COCKCROFT-GAULT EQUATION TO CALCULATE CrCl

Cockcroft-Gault Equation to calculate creatinine Clearance (CrCl):



$$\text{CrCl} = [140 - \text{age (years)}] \times \text{weight (kg)} / [\text{serum creatinine (mg/dL)} \times 72] \times 0.85 \text{ (if female)}$$

OR

$$\text{CrCl (mL/min)} = \frac{(140 - \text{age}) \times 1.2 \times \text{body weight (kg)}}{\text{serum creatinine } (\mu\text{mol/L})} \times 0.85 \text{ (ONLY if female)}$$

- Use IBW *unless* patient is obese (30% over their IBW) or the patient is less than their IBW
- Less than IBW: ABW should be used instead
- >30%* over IBW: AdjBW should be used instead

* "30% over IBW" is equivalent to "130% of IBW"

$$\text{IBW (males)} = 50 \text{ kg} + (2.3 \times \text{every inch over 5 feet})$$

$$\text{IBW (females)} = 45.5 \text{ kg} + (2.3 \times \text{every inch over 5 feet})$$

$$\text{AdjBW} = \text{IBW} + 0.4 (\text{ABW} - \text{IBW})$$

POSSIBLE DRUG CAUSES

Not an exhaustive list!

Pre-renal	Intrinsic/Renal		Post-renal
	Tubular	Interstitial	Crystal induced
ACEi/ARBs	Aminoglycosides	Penicillin analogues	Acyclovir
NSAIDs	Radiographic contrast media	Cephalosporins	Indinavir
Cyclosporine	Cisplatin	Ciprofloxacin	Sulfonamides
Tacrolimus	Methotrexate	Acyclovir	
Diuretics	Ethylene glycol	Interferon	
Laxatives	Amphotericin B	PPIs	
SGLT2i	Lithium	Allopurinol	
	Vancomycin	NSAIDs	

GOALS OF THERAPY

- ✓ Identify the underlying cause of AKI
- ✓ Treat life-threatening conditions
 - ✓ Infection
 - ✓ Hyperkalemia and other electrolyte abnormalities
 - ✓ Hypovolemia
 - ✓ Acid-base disorder
- ✓ Treat reversible causes
- ✓ Remove nephrotoxins
- ✓ Hydrate kidneys
- ✓ Remove obstruction
- ✓ Prevent mortality associated with AKI

TREATMENT

PRE-RENAL & INTRINSIC AKI

- Fluid resuscitation with isotonic crystalloid solutions (Ringer's Lactate preferred over normal saline)
 - Monitor fluid balance (input & output)
- Sodium bicarbonate IV for metabolic acidosis and alkalinizing urine (rhabdomyolysis)
- Vasopressors and fluids in shock with/at risk of AKI
- Diuretics for treatment of fluid overload
- Correct electrolyte imbalances ($\uparrow K^+$ $\uparrow PO_4^{3-}$ $\uparrow Mg^{2+}$)
- Remove offending agents

TREATMENT

POST-RENAL AKI

- Remove whatever is causing the obstruction
- Renal replacement therapy or dialysis to remove fluid or toxins
 - Refractory hyperkalemia (despite treatment)
 - Metabolic acidosis
 - Volume overload despite diuretic use
 - Pulmonary edema
 - Mental status changes/uremia (seizure, loss of consciousness, bleeding)
 - Rhabdomyolysis

NON-PHARMACOLOGICAL TREATMENT

- Electronic alert systems in acute care settings can allow for early detection of AKI
- Monitoring of patients taking nephrotoxic drugs
- KDIGO guidelines recommend caloric intake of 20-30 kcal/kg/day regardless of the stage of kidney impairment
 - Preferably oral administration

PREVENTION

IDENTIFY PATIENTS AT HIGH RISK FOR AKI AND TREAT ACCORDING TO RISK FACTORS

Contrast-Induced AKI

- Use IV contrast only if necessary, minimize contrast volume, low-osmolar (non-ionic) contrast
- Pre- and post-procedure prophylaxis IV Fluids (ex. NS 0.9%, 1-1.5 mL/kg/h x 3-12 hours before procedure and 6-12 hours post-procedure)
- Can also use sodium bicarbonate and N-acetylcysteine (weak evidence, uncommonly used)

Cancer chemotherapy with risk of TLS

- Hydration (IV fluids)
- Allopurinol (e.g. 100 mg/m²/dose PO TID with max. 800 mg/day) 1-2 days before the start of chemotherapy to prevent uric acid nephropathy
- Rasburicase (e.g. 0.2mg/kg/day IV infused over 30 minutes) for 3-5 days (up to 7 days). Chemo may be initiated as soon as four hours after first dose.

PREVENTION

IDENTIFY **PATIENTS AT HIGH RISK FOR AKI** AND TREAT ACCORDING TO RISK FACTORS

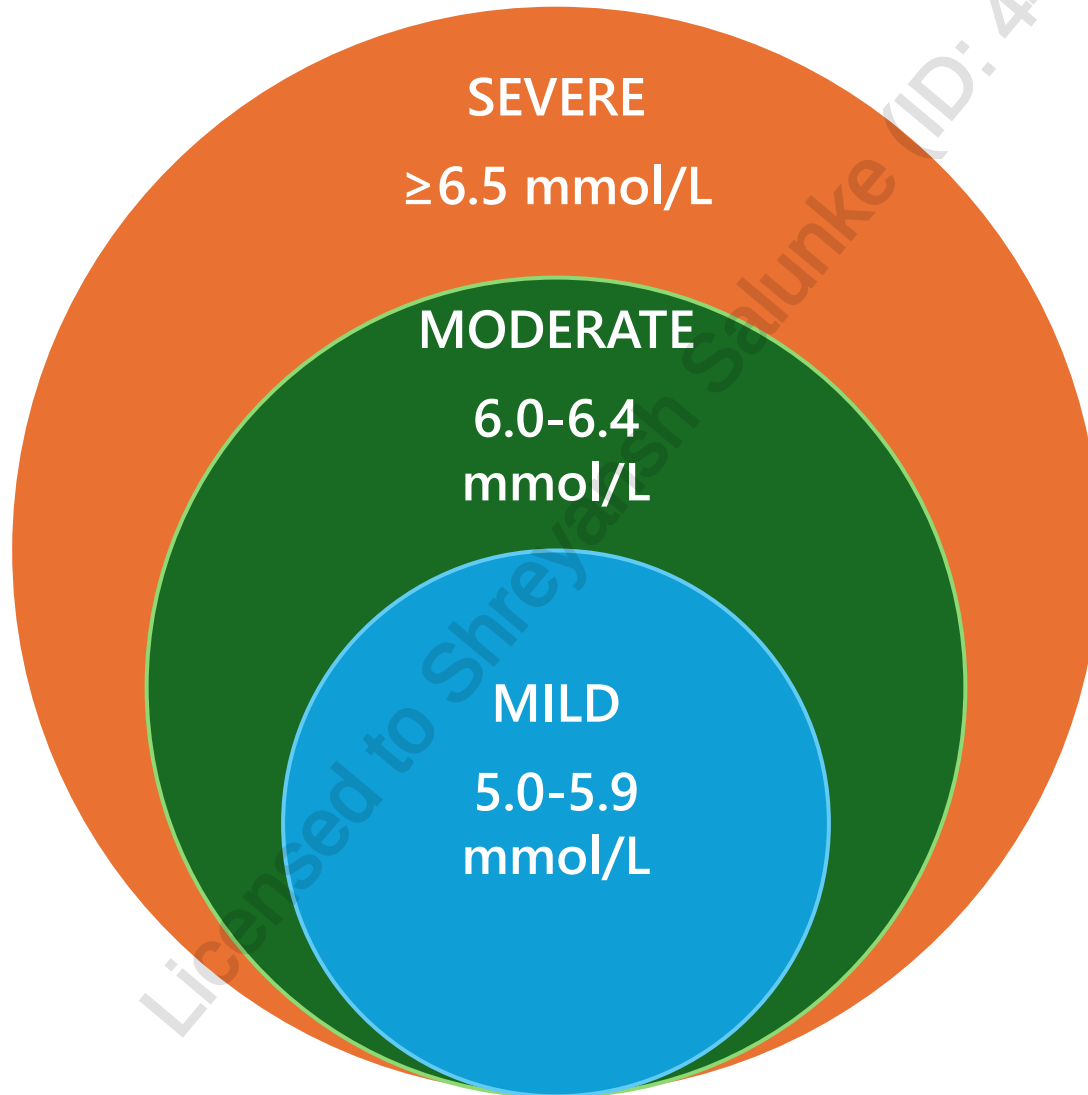
Nephrotoxic medication exposure

- **Avoid** if possible
- Therapeutic **drug monitoring** when available
- **Appropriate dose, interval** (e.g. aminoglycosides once daily vs Q8H), duration and formulation (e.g. liposomal amphotericin B)
- **Preventative hydration** (e.g. cyclophosphamide)

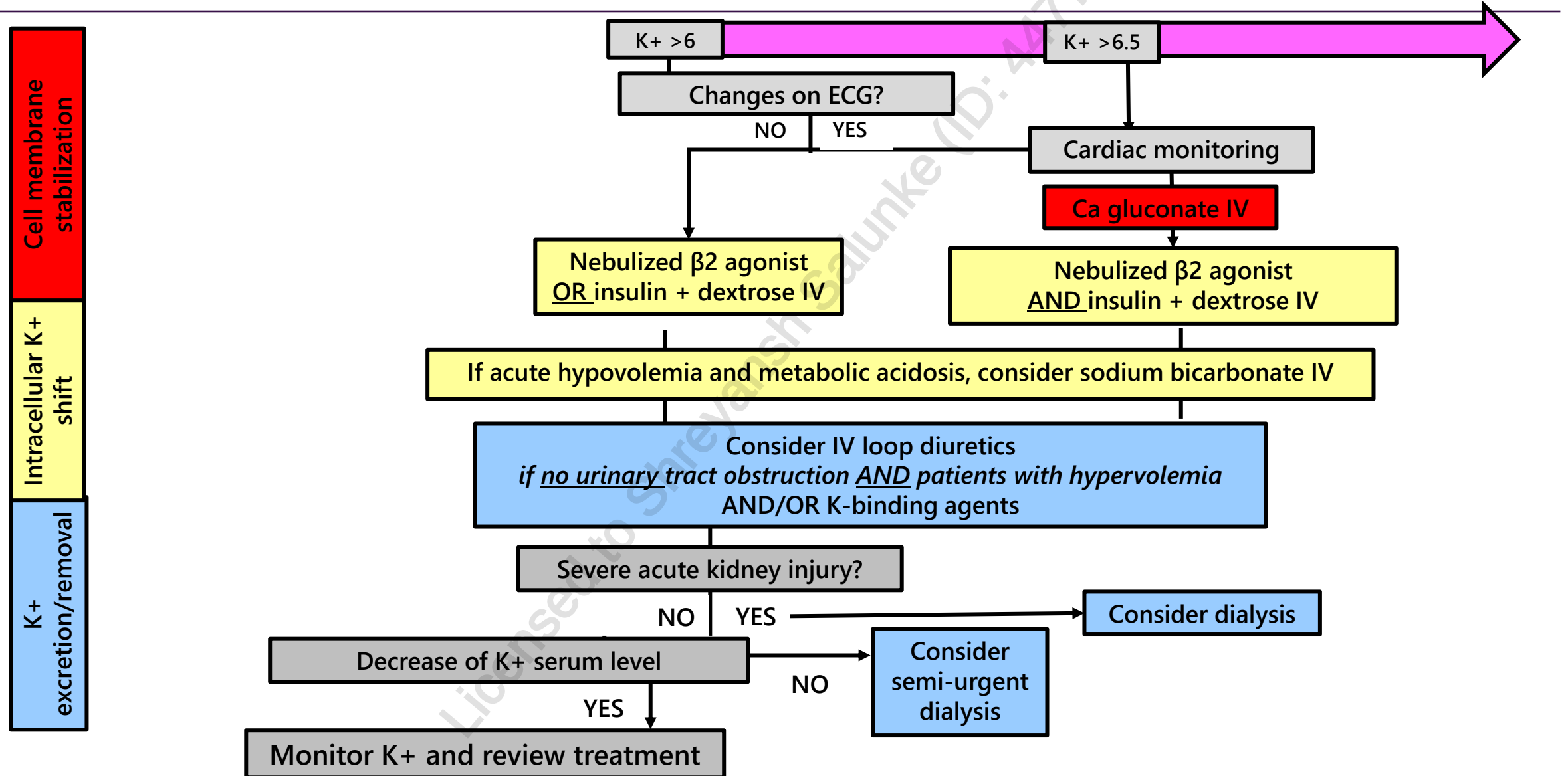
Peri-operative

- **Volume resuscitation/prevention of hypotension**
- Consider **holding RAAS antagonist and SGLT2i**

HYPERKALEMIA



MANAGEMENT OF HYPERKALEMIA



PHARMACOLOGIC TREATMENT FOR HYPERKALEMIA

Agent	Mechanism	Onset	Duration	Dose	Complications
Calcium gluconate IV	Membrane stabilization Use if ECG changes	Immediate	30-60 min	1 g IV bolus	Hypercalcemia
Sodium bicarbonate	Shift of K ⁺ into cells Use if acidosis	1-2 hr	Duration of infusion	150 mmol/L IV variable rate	Metabolic alkalosis (severe)
Insulin (regular)	Shift of K ⁺ into cells	15 min	4-6 hr	10 units IV push with 25-40 g of 50% dextrose	Hypoglycemia (especially if not enough dextrose given)
Salbutamol	Shift of K ⁺ into cells	30 min	2 hr	20 mg nebulized	Transient tachycardia
Furosemide IV	Eliminate K ⁺	15 min	2-3 hr	40-80 mg IV bolus	Hypotension due to volume depletion
Sodium polystyrene sulfonate	Eliminate K ⁺	1.5-24 hr	4-6 hr	15-30 g in 15-30 mL 70% sorbitol orally <u>OR</u> 30-50 g in 150- 200 mL water rectally	Do not use sorbitol rectally Sorbitol 70% may cause bowel ischemia
Dialysis	Eliminate K ⁺	Immediate	Duration of dialysis	Based on BP, fluid status	Hypotension

Adapted from: Weisberg , L. "Management of Severe Hyperkalemia," Crit Care Med 2008 Vol. 36, No. 12

MONITORING PARAMETERS

- Renal function
 - Creatinine
 - Urea
 - Urine output
- Fluid balance
 - Input and output on an hourly basis
 - Blood pressure
- pH
- Electrolytes (K^+ , Na^+ , Ca^{2+})
- Improvement of signs & symptoms

RENAL DOSE ADJUSTMENTS

- Renally adjust medications in CKD and other cases of decreased kidney function to decrease the risk of the patient also suffering from an AKI
- Medications will also need to be adjusted while the patient has an AKI or renal failure
- Useful Resources for renal adjustments of drugs
 - Bennett's Drug Prescribing in Renal failure
 - Renal Drug Handbook (3rd Ed)
 - UpToDate® LexiDrug (formerly Lexicomp®)
 - Louisville Kidney Disease Program

TAKEAWAY

- Assess the cause of renal dysfunction (pre-, intra- or post-renal)
- Identify and correct any drug or medical causes
- Stop nephrotoxins
- Remove obstruction
- Prevention strategies
 - Avoid nephrotoxins
 - Maintain hydration and renal perfusion
 - Use pre- and post-procedure prophylaxis for the prevention of contrast-related AKI
- Correct hypovolemia and electrolyte abnormalities – to decrease the risk of arrhythmias
- Give IV diuretics to produce diuresis only after hypovolemia corrected
- Dialysis based on appropriate indications

CASE 1

CC is a 74-year-old female who presents to the rapid assessment clinic. She has been experiencing increasing weakness over the last few days, poor balance and cognitive decline. Blood is drawn and a urine sample is collected for analysis. She is a former smoker and her past medical history is significant for metastatic renal cell carcinoma (diagnosed 1.5 years ago), chronic kidney disease (CKD) (diagnosed 6 months ago), hypothyroidism (new diagnosis), gastroesophageal reflux disease (GERD, diagnosed 1 year ago), and hypertension (diagnosed 5 years ago). She is currently taking levothyroxine 0.025 mg PO daily, pantoprazole 40 mg PO daily, ramipril 10 mg PO daily, melatonin 5 mg PO nightly, and she started on nivolumab 240 mg IV Q2 weeks (initiated 2 months ago). She is allergic to contrast media.

Objective findings:

- SCr = now - 144, 24h ago - 93 (60-110 $\mu\text{mol/L}$)
- Na = 134 (135-150 mmol/L)
- K⁺ = 5.3 (3.5-5 mmol/L)
- HCO₃⁻ = 21 (21-32 mmol/L)
- Urine: + WBC, protein
- BP = 98/57 mmHg
- HR = 84 beats per minute



CASE 1 - QUESTIONS

1. What are CC's risk factors for AKI? Select the MOST accurate answer.
 - a) CKD, hypertension, renal cell carcinoma, age
 - b) CKD, hypertension, hypothyroidism, age, smoking
 - c) CKD, hypertension, renal cell carcinoma, smoking
 - d) CKD, hypertension, renal cell carcinoma, medications, age

2. Which is NOT a possible cause of CC's AKI?
 - a) Pre-renal volume depletion
 - b) Intrinsic medication-related due to nivolumab
 - c) Post-renal due to tumour-related obstruction
 - d) Intrinsic medication-related due to levothyroxine

CASE 1 - QUESTIONS

3. A renal ultrasound is completed and shows that there is no hydronephrosis. The ultrasound does show a heterogeneous mass in the left kidney, but no obstructed ureters. Which of the following causes of AKI can now be ruled out?
- a) Pre-renal due to hypovolemia
 - b) Intrinsic damage due to renal mass
 - c) Intrinsic damage due to nivolumab
 - d) Post renal obstruction due to renal mass
4. Which of the following medications would be effective at removing K⁺ from the body?
- a) Sodium bicarbonate
 - b) Insulin
 - c) Sodium polystyrene sulfonate
 - d) Salbutamol

CASE 2

RR is a 65-year-old man who presents to the emergency room with nausea, vomiting, fatigue, muscle aches, fever and burning with urination. On presentation, he is breathless, pale and diaphoretic. RR was feeling well until recently when he developed a gout flare. He had some leftover indomethacin from a previous gout flare and self-treated with 50 mg PO TID for the last 2 days. Physical exam reveals elevated JVP by 2 cm, HR = 60 beats per minute, and BP=156/94 mmHg. During the exam, his level of consciousness begins to decline. His past medical history includes hypertension, type 2 diabetes mellitus and gout. He is currently taking hydrochlorothiazide 25 mg PO daily, ramipril 5 mg PO daily, and empagliflozin 25 mg PO daily (initiated last week).

Labs:

- pH = 7.26 (7.36-7.4)
- SCr = 286 (60-110 $\mu\text{mol/L}$)
- Na^+ = 138 (135-150 mmol/L)
- K^+ = 7.0 (3.5-5 mmol/L)
- HCO_3^- = 18 mEq/L (21-32 mmol/L)
- Ca^{+} = 2.1 (2.2-2.5 mmol/L)
- PO_4 = 1.7 (0.9-1.5 mmol/L)
- Glucose = 10.5 mmol/L



CASE 2 - QUESTIONS

1. What is the most concerning aspect of RR's presentation and why?
 - a) Tachycardia and volume loss with risk of hemodynamic shock
 - b) Elevated potassium with risk of leading to arrhythmia
 - c) Volume overload with risk of heart failure
 - d) Elevated blood glucose resulting in hyperglycemia

2. Choose the next best step for RR's management:
 - a) Obtain ECG, obtain renal ultrasound, start dialysis
 - b) Obtain renal ultrasound, give IV 0.9% NS, start antibiotics
 - c) Conduct cardiac monitoring, give calcium gluconate IV, correct potassium, start ceftriaxone IV
 - d) Correct potassium, start IV furosemide, start ceftriaxone IV

CASE 2 - QUESTIONS

3. What are the precipitating factors of AKI for RR?
- a) Indomethacin, diabetes mellitus, hypertension, possible urinary tract infection
 - b) Diabetes mellitus, hypertension, indomethacin use, hydrochlorothiazide, empagliflozin
 - c) Indomethacin, hydrochlorothiazide, possible urinary tract infection
 - d) Indomethacin, empagliflozin, hydrochlorothiazide, possible urinary tract infection
4. An ECG is done and shows peaked T waves and a shortened QT interval. One dose of calcium gluconate IV has already been given. What is the most appropriate next step for treating RR's hyperkalemia?
- a) Another calcium gluconate 1 g IV bolus, hold contributing medications, monitor K+
 - b) Insulin + dextrose 50%, nebulized salbutamol, hold ramipril, hydrochlorothiazide
 - c) Sodium bicarbonate PO, insulin + dextrose 50%, furosemide IV
 - d) Sodium polystyrene sulfate enema, nebulized salbutamol, furosemide PO

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CHANGE LOG

July 2025

- Formatting, grammar changes throughout; legend updated
- Slide 6: Changed “gender” to “sex”
- Slide 23: Updated dose of allopurinol for TLS
- Slide 30: Added nivolumab dose to case

November 2025

- Slide 17: Changed eGFR to CrCl, removed ‘ideal’ from equation, added alternative equation and added a note to clarify that 30% and 130% are equivalent

January 2025

- Slide 33: Changed ‘no unobstructed ureters’ to ‘no obstructed ureters’

February 2026

- Slide 6: Added rhabdomyolysis as risk factor, contrast dye, and removed male sex
- Slide 18: Added SGLT2i
- Slide 20: Ringer’s preferred over NS
- Slide 23: Added Rasburicase
- Slide 24: added SGLT2i to hold list